

Project Development Phase

Code-Layout, Readability and Reusability

Date	10/07/2026
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Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the **Credit Card Approval Prediction** project's codebase in terms of structure, readability, maintainability, and reusability. The project follows good programming practices by maintaining a modular architecture, meaningful naming conventions, proper documentation, and reusable components. These practices make the project easier to understand, debug, and extend with additional machine learning models or datasets.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation maintained throughout the project.	Yes	Improves code readability.
2	Proper File Structure	Source code, datasets, models, and notebooks are organized into separate folders.	Yes	Easy to navigate and maintain.
3	Meaningful Variable Names	Variables such as age, income, credit_score, and prediction clearly represent their purpose.	Yes	Enhances code understanding.
4	Function / Method Names	Functions like load_data(), preprocess_data(), train_model(), and predict_approval() are descriptive.	Yes	Makes code self-explanatory.
5	Code Comments	Important preprocessing steps and model logic are documented using comments.	Yes	Simplifies maintenance.
6	Modular Design	Data preprocessing, training, evaluation, and prediction are implemented in separate modules/functions.	Yes	Promotes code reuse.
7	No Redundant Code	Duplicate code is avoided by using reusable functions.	Yes	Improves efficiency.
8	Error Handling	Input validation and exception handling are implemented for data	Yes	Prevents runtime failures.

		loading and prediction.		
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module	Python (Pandas, NumPy)	Data cleaning and feature preparation	High
2	Feature Engineering Module	Python (Scikit-learn)	Feature scaling and encoding	High
3	Model Training Module	Python (Scikit-learn)	Training multiple ML algorithms	High
4	Prediction Module	Python	Predicting credit card approval	High
5	Evaluation Module	Python (Scikit-learn)	Accuracy, Precision, Recall, F1-Score, ROC-AUC evaluation	High

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized project with separate modules and folders.
Readability	5	Clear variable names, proper formatting, and descriptive functions.
Reusability	5	Modular functions can be reused for similar ML classification projects.
Documentation / Comments	5	Adequate comments and documentation provided throughout the code.
Overall Score	5 / 5	Excellent code quality with maintainable and reusable architecture.